



GIT

Introdução

PAULO R T CÂNDIDO

Git – Comandos Principais

mkdir testeGit

Transformar pasta em um repositório git

git init

Mostrar o nome do branch em que se está no momento

git branch --show-current

Renomeia branch para main

git branch -m main

Listar as configurações globais

git config --global --list

Consultar configurações globais

git config --global user.email

Alterar configurações globais

git config --global user.name "Paulo Candido"

Git – Comandos Principais

Configurar repositório remoto no GITHUB (necessário criar manualmente no Github)

```
git remote add origin https://github.com/prtcandido/testeGit.git
```

Verificar configuração de armazenamento remoto

```
git remote -v
```

Alterar URL de armazenamento remoto

```
git remote set-url origin https://github.com/prtcandido/gitaula1.git
```

Remover armazenamento remoto

```
git remote remove origin
```

Reconfigurar <<<<

```
git remote add origin https://github.com/prtcandido/testeGit.git
```

Git – Comandos Principais

Criar estes arquivos

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ cat a.log
teste
teste
x
```

Listar os arquivos cujo versionamento está sendo controlado

```
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    a.log
    a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Criar arquivo .gitignore, todos os arquivos log serão ignorados
nano .gitignore

```
GNU nano 7.2
*.log
```

Todos os arquivos log são ignorados

```
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .gitignore
    a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Adicionar arquivo no stage para commit

```
prtcandido@PauloAcer:~/testeGit$ git add a.txt
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   a.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
```

Retirar arquivo do stage

```
prtcandido@PauloAcer:~/testeGit$ git rm --cached a.txt
rm 'a.txt'
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        a.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Git – Comandos Principais

Adiciona todos os arquivos novos e modificados no stage para commit (pacote de alterações)

```
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   a.txt

prtcandido@PauloAcer:~/testeGit$ git commit -m primeiro
[main (root-commit) 6f4b108] primeiro
 2 files changed, 5 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 a.txt
```

Listar commits realizados

```
prtcandido@PauloAcer:~/testeGit$ git log --oneline
6f4b108 (HEAD -> main) primeiro
```

Git – Comandos Principais

Primeiro push para remoto origin do branch main (necessário criar repositório no github)

```
prtcandido@PauloAcer:~/testeGit$ git push -u origin main
Username for 'https://github.com': prtcandido
Password for 'https://prtcandido@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 283 bytes | 283.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/prtcandido/testeGit.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
prtcandido@PauloAcer:~/testeGit$ |
```

Diferença de conteúdo entra a versão atual de a.txt e a versão do último commit.

```
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praiaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git diff a.txt
diff --git a/a.txt b/a.txt
index 7ce909d..8132ede 100644
--- a/a.txt
+++ b/a.txt
@@ -1,4 +1,4 @@
 Fatec
-Praia
+Praiaaaa
 Grande
```

Diferença com o arquivo do commit 6f4b108

```
prtcandido@PauloAcer:~/testeGit$ git diff 6f4b108 a.txt
diff --git a/a.txt b/a.txt
index 7ce909d..8132ede 100644
--- a/a.txt
+++ b/a.txt
@@ -1,4 +1,4 @@
 Fatec
-Praia
+Praiaaaa
 Grande
```


Git – Comandos Principais

Recupera versão anterior de um commit (pacote de alterações) específico.

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praiaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git restore --source 6f4b108 a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ |
```

Comparar com arquivo que está em stage

git diff --cached a.txt

Git – Comandos Principais

Criar dois branches (cópias!!!): b1 baseado na main; b2 baseado no commit 6f4b108

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git branch b1
prtcandido@PauloAcer:~/testeGit$ git checkout -b b2 6f4b108
Switched to a new branch 'b2'
```

Modificar o arquivo a.txt de forma diferente nos dois branches e fazer commit das mudanças.

```
prtcandido@PauloAcer:~/testeGit$ git checkout b1
Switched to branch 'b1'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b1_1
[b1 6a79844] b1_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout b2
Switched to branch 'b2'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatec
Praia
Grandeeee

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b2_1
[b2 c95e662] b2_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ |
```

Git – Comandos Principais

Mesclar a alterações feitas nos branches b1 e b2 no main.

```
prtcandido@PauloAcer:~/testeGit$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
prtcandido@PauloAcer:~/testeGit$ git merge b1
Updating 6f4b108..6a79844
Fast-forward
 a.txt | 2 + -
 1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande

prtcandido@PauloAcer:~/testeGit$ git merge b2
Auto-merging a.txt
Merge made by the 'ort' strategy.
 a.txt | 2 + -
 1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praia
Grande
```

Git – Comandos Principais

Gerar conflito para merge alterando a mesma linha de a.txt nos branches main e b1

```
prtcandido@PauloAcer:~/testeGit$ git branch --show-current
main
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaxxxxxxxxx
Grandegee

prtcandido@PauloAcer:~/testeGit$ git checkout b1
error: Your local changes to the following files would be overwritten by checkout:
a.txt
Please commit your changes or stash them before you switch branches.
Aborting
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m main_1
[main e7e666e] main_1
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout b1
Switched to branch 'b1'
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaaaaaaaaaa
Grande

prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b1_2
[b1 df185ca] b1_2
1 file changed, 1 insertion(+), 1 deletion(-)
prtcandido@PauloAcer:~/testeGit$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)
prtcandido@PauloAcer:~/testeGit$ git merge b1
Auto-merging a.txt
CONFLICT (content): Merge conflict in a.txt
Automatic merge failed; fix conflicts and then commit the result.
prtcandido@PauloAcer:~/testeGit$ |
```

```
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
<<<<<< HEAD
Praiaxxxxxxxxx
Grandegee
=====
Praiaaaaaaaaaa
Grande
>>>>>> b1
```

ours VS theirs

Se houver um conflito de código, você diz ao Git quem deve "vencer" automaticamente.

- git merge -Xours feature: Em caso de conflito, mantém a versão da sua branch atual (**main**).
- git merge -Xtheirs feature: Em caso de conflito, aceita as mudanças da branch que está vindo (**feature**).

Git – Comandos Principais

```
prtcandido@PauloAcer:~/testeGit$ git branch -a
b1
b2
* main
remotes/origin/main
prtcandido@PauloAcer:~/testeGit$ git checkout -b b3
Switched to a new branch 'b3'
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc
Praiaxxxxxxxxx
Grandegee
prtcandido@PauloAcer:~/testeGit$ nano a.txt
prtcandido@PauloAcer:~/testeGit$ cat a.txt
Fatecccc#####
Praiaxxxxxxxx$$$$$$$$$$$$$$$
Grandegee@@@@@@@@@@@@@@@@
prtcandido@PauloAcer:~/testeGit$ git add .
prtcandido@PauloAcer:~/testeGit$ git commit -m b3_1
[b3 5ccf9a0] b3_1
1 file changed, 3 insertions(+), 3 deletions(-)
prtcandido@PauloAcer:~/testeGit$ git push -u origin b3
Username for 'https://github.com': prtcandido
Password for 'https://prtcandido@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 306 bytes | 306.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'b3' on GitHub by visiting:
remote: https://github.com/prtcandido/testeGit/pull/new/b3
remote:
To https://github.com/prtcandido/testeGit.git
* [new branch]      b3 -> b3
branch 'b3' set up to track 'origin/b3'.
prtcandido@PauloAcer:~/testeGit$
```

Feito push do novo Branch (b3) é possível gerar um Pull Request no Github

Git – Pull Request

Open a pull request

Create a new pull request by comparing changes across two branches. If you need to, you can also [compare across forks](#). [Learn more about diff comparisons here](#).

 base: main  compare: b3 ✓ Able to merge. These branches can be automatically merged.



Add a title *
Aula sobre GIT 

Add a description

Write Preview

H B I        @   

Apenas um teste

 Markdown is supported  Paste, drop, or click to add files

Reviewers 
No reviews

Assignees 
No one—[assign yourself](#)

Labels 
None yet

Projects 
None yet

Milestone 
No milestone

Development
Use [Closing keywords](#) in the description to automatically close issues

Helpful resources

Create pull request 

Git – Pull Request

Aula sobre GIT #1 ✎

✓ Ready to merge

Code ▾

🔗 Open prtcandido wants to merge 1 commit into `main` from `b3` 📄

💬 Conversation 0

🔑 Commits 1

📄 Checks 0

📄 Files changed 1

+3 -3 🟢🔴🔴🔴



prtcandido commented 1 minute ago

Owner ⋮

Apenas um teste



5ccf9a0

Pronto para fazer o Merge



No conflicts with base branch

Merging can be performed automatically.

Merge pull request ▾

You can also merge this with the command line. [View command line instructions.](#)

Still in progress? [Convert to draft](#)



Add a comment

Write

Preview

H B I ☰ <> 🔗 ☰ ☰ ☰ 📎 @ 📄 ↩️ 🗑️

Add your comment here...

Reviewers



No reviews

Still in progress? [Convert to draft](#)

Assignees



No one—[assign yourself](#)

Labels



None yet

Projects



None yet

Milestone



No milestone

Development



Successfully merging this pull request may close these issues.

None yet

Notifications

[Customize](#)

Git – Pull Request

https://github.com/prtcandido/testeGit/settings/branches

prtcandido / testeGit

Code Issues Pull requests 1 Actions Projects Wiki Security Insights **Settings**

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models Preview

Webhooks

Branch protection rules

Classic branch protections have not been configured
Define branch rules to disable force pushing, prevent branches from being deleted, or require pull requests b
Learn more about [repository rules](#) and [protected branches](#).

Add branch ruleset **Add classic branch protection rule**

Git – Pull Request

General

Access

- Collaborators
- Moderation options

Code and automation

- Branches**
- Tags
- Rules
- Actions
- Models
- Webhooks
- Copilot
- Environments
- Codespaces
- Pages

Security

- Advanced Security
- Deploy keys
- Secrets and variables

Integrations

Branch protection rule

Protect your most important branches

[Branch protection rules](#) define whether collaborators can delete or force push to the branch and set requirements for any pushes to the branch, such as passing status checks or a linear commit history.

[Your GitHub Free plan](#) can only enforce rules on its public repositories, like this one.

Branch name pattern *

main

Protect matching branches

- ☒ **Require a pull request before merging**
When enabled, all commits must be made to a non-protected branch and submitted via a pull request before they can be merged into a branch that matches this rule.
- ☒ **Require approvals**
When enabled, pull requests targeting a matching branch require a number of approvals and no changes requested before they can be merged.
Required number of approvals before merging: 1
- ☐ **Dismiss stale pull request approvals when new commits are pushed**
New reviewable commits pushed to a matching branch will dismiss pull request review approvals.
- ☐ **Require review from Code Owners**
Require an approved review in pull requests including files with a designated code owner.
- ☐ **Require approval of the most recent reviewable push**

Git – Pull Request

The screenshot shows a GitHub Pull Request (PR) page for the repository 'prtcandido / testeGit'. The PR is titled 'Aula sobre GIT #1' and is in the 'Awaiting approval' state. It shows that 'prtcandido' wants to merge 1 commit into the 'main' branch from the 'b3' branch. The PR has 1 conversation, 1 commit, 0 checks, and 1 file changed. The main comment from 'prtcandido' (the owner) says 'Apenas um teste' and was made 'last week'. The PR is currently blocked from merging because it requires at least 1 approving review. The 'Reviewers' section on the right shows no reviews, with a message: 'No reviews—at least 1 approving review is required. Still in progress? [Convert to draft](#)'. The 'Assignees' section shows 'prtcandido' is assigned. The 'Labels' section shows 'None yet'. The 'Projects' section shows 'None yet'. The 'Milestone' section shows 'No milestone'. The 'Development' section shows 'Development'. At the bottom, there are two error messages: 'Review required' and 'Merging is blocked', both stating 'At least 1 approving review is required by reviewers with write access.'.

https://github.com/prtcandido/testeGit/pull/1

prtcandido / testeGit

de Issues Pull requests 1 Actions Projects Wiki Security Insights Settings

Aula sobre GIT #1

Awaiting approval Code

Open prtcandido wants to merge 1 commit into main from b3

Conversation 0 Commits 1 Checks 0 Files changed 1 +3 -3

prtcandido commented last week

Apenas um teste

prtcandido self-assigned this 20 minutes ago

Review required
At least 1 approving review is required by reviewers with write access.

Merging is blocked
At least 1 approving review is required by reviewers with write access.

Reviewers
No reviews—at least 1 approving review is required.
Still in progress? [Convert to draft](#)

Assignees
prtcandido

Labels
None yet

Projects
None yet

Milestone
No milestone

Development

Git – Pull Request

Aula sobre GIT #1

● Awaiting review

 Open [prtcandido](#) wants to merge 1 commit into `main` from `b3` 

 Conversation 0  Commits 1  Checks 0  Files changed 1



[prtcandido](#) commented [last week](#)

Owner

...

Apenas um teste



 [prtcandido](#) self-assigned this [31 minutes ago](#)

 [prtcandido](#) added the `bug` label [4 minutes ago](#)

Em **CI/CD e Automação**, as labels são gatilhos. Por exemplo, se um Pull Request recebe a label `deploy-to-prod`, um workflow pode disparar o deploy automaticamente.

[5ccf9a0](#)

Reviewers

No reviews—at least 1 approved review is required to merge
Still in progress? [Convert to draft](#)

Assignees

 [prtcandido](#)

Labels

`bug`

Projects

None yet